

Econ 139: Labor Economics

Technology: Automation Part 2

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Review

- Last class studied how exposure to different automation technologies (most clearly industrial robots) has impacted labor-market outcomes
 - Found it is responsible for (1) large share of the decline in the labor share and (2) large share of the increase in inequality
- Here are a few conceptual questions about the last two classes that you might find on the final

Review

- We studied two different papers on industrial robots. Pick one of the two and describe the identification strategy: In other words, how did they identify whether robots increase or decrease jobs
 - **Robots and Jobs: Evidence from US Labor Markets** by Acemoglu and Restrepo
 - **Robot Adoption and Labor Market Dynamics** by Humlum
- Describe one main result from the paper? You do not need to have specific numbers, but instead state an overall clear takeaway. Describe why (theoretically) the result may have been different.

Review

- Dekle (2020) finds that in Japan (which uses about 10x the amount of robots as the US), robots have had a clear displacement effect. They have replaced humans in certain tasks. Regardless, Dekle finds that robots have led to an **increase** in the demand for workers overall. Describe how this is possible.
- Imagine Robot A and Robot B perform the exact same tasks. However, Robot A is much more efficient at the task and can greatly boost productivity, while Robot B is only marginally more efficient than a standard worker. Which robot would you expect to have a more harmful impact on workers?

AI

- In this lecture, we are going to discuss some recent work on how AI has impacted the labor market
 - Given this is a relatively recent technology, we are still at the very early stages of understanding how this will impact the labor market generally
 - While pace of work is rapid, there are many things we don't know and need time to see how things evolve

What is AI?

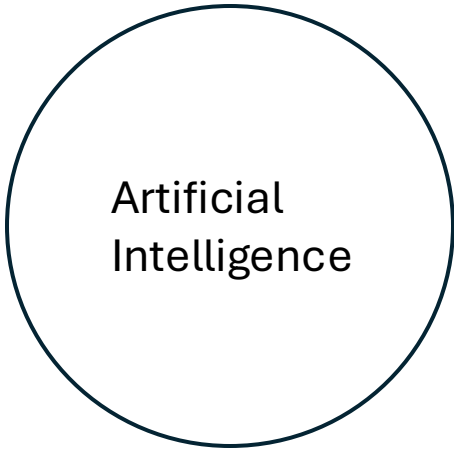
- Artificial Intelligence has been around for a long time
- The birthplace of AI is often thought to be a 1956 Conference at Dartmouth
- Here is a snippet of the conference proposal

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The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves

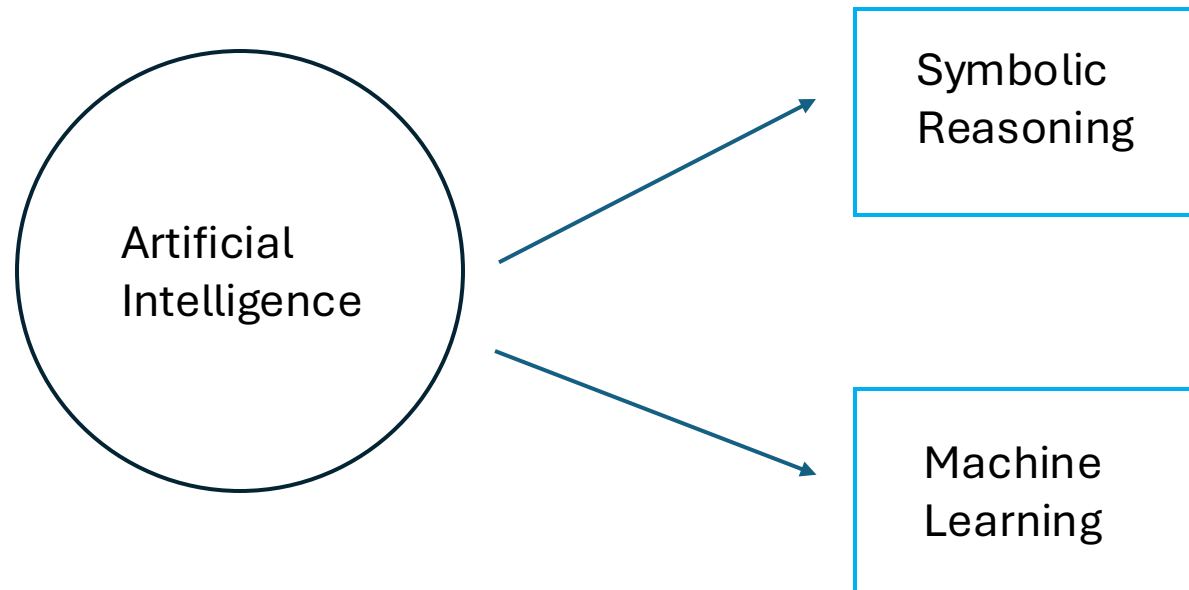
AI Approaches



Artificial
Intelligence

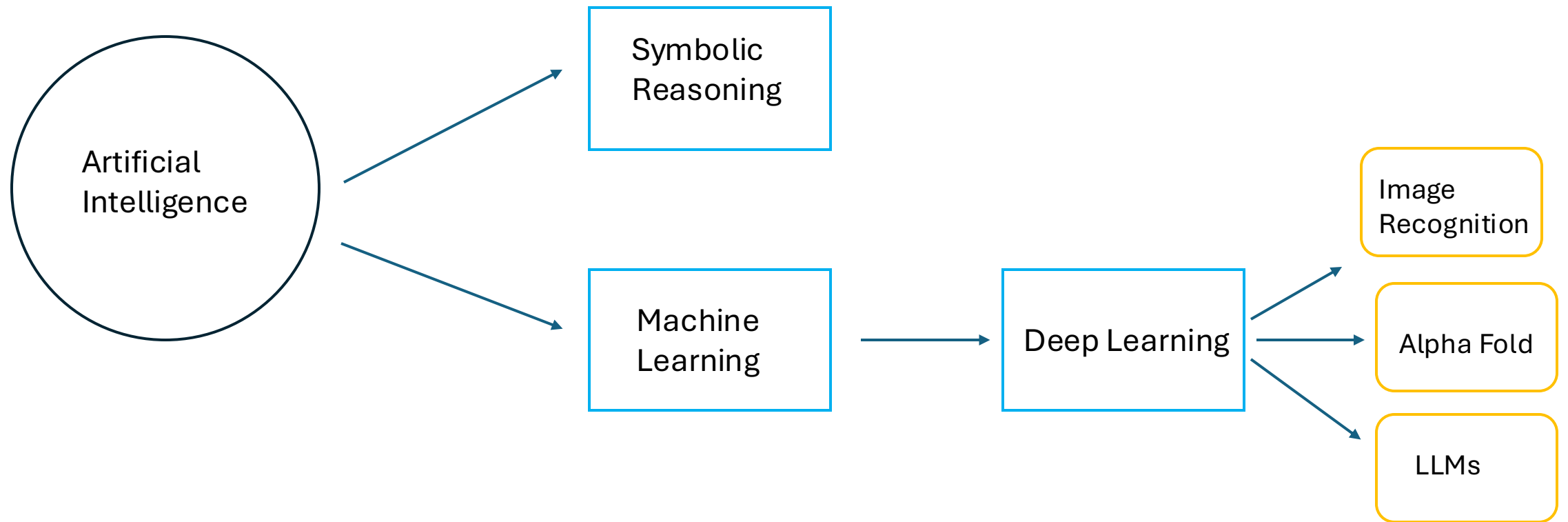
AI Approaches

Dominated early period of Artificial Intelligence research

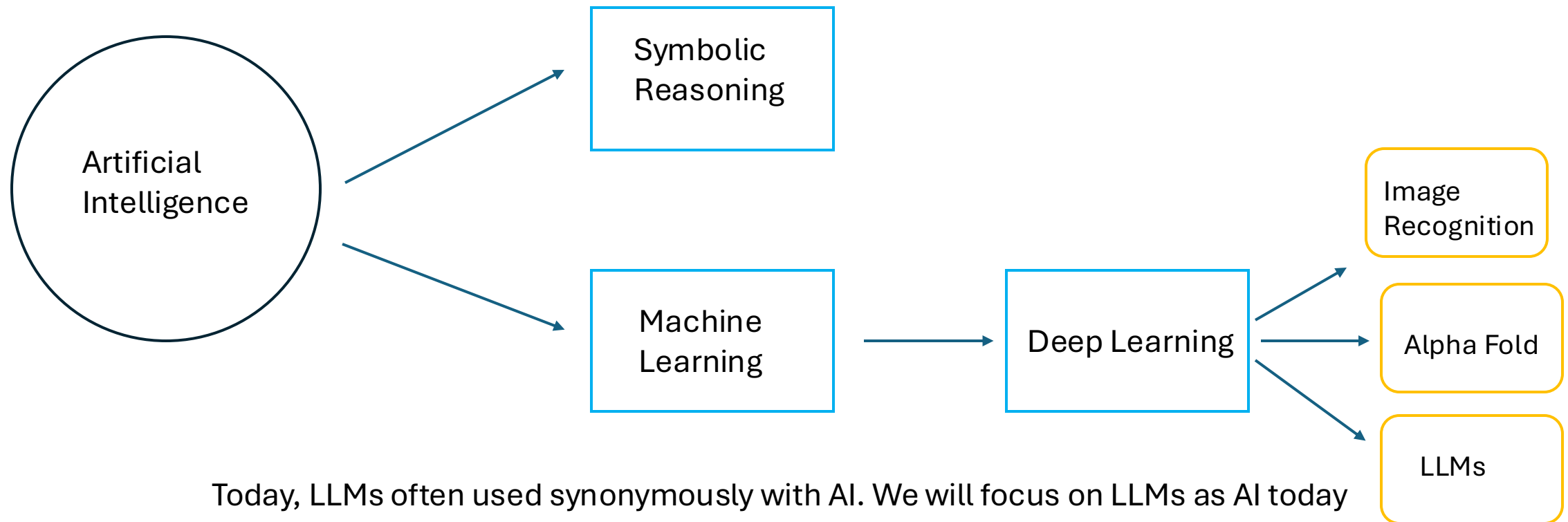


First suggested in 1950s, but poor performance overall

AI Approaches



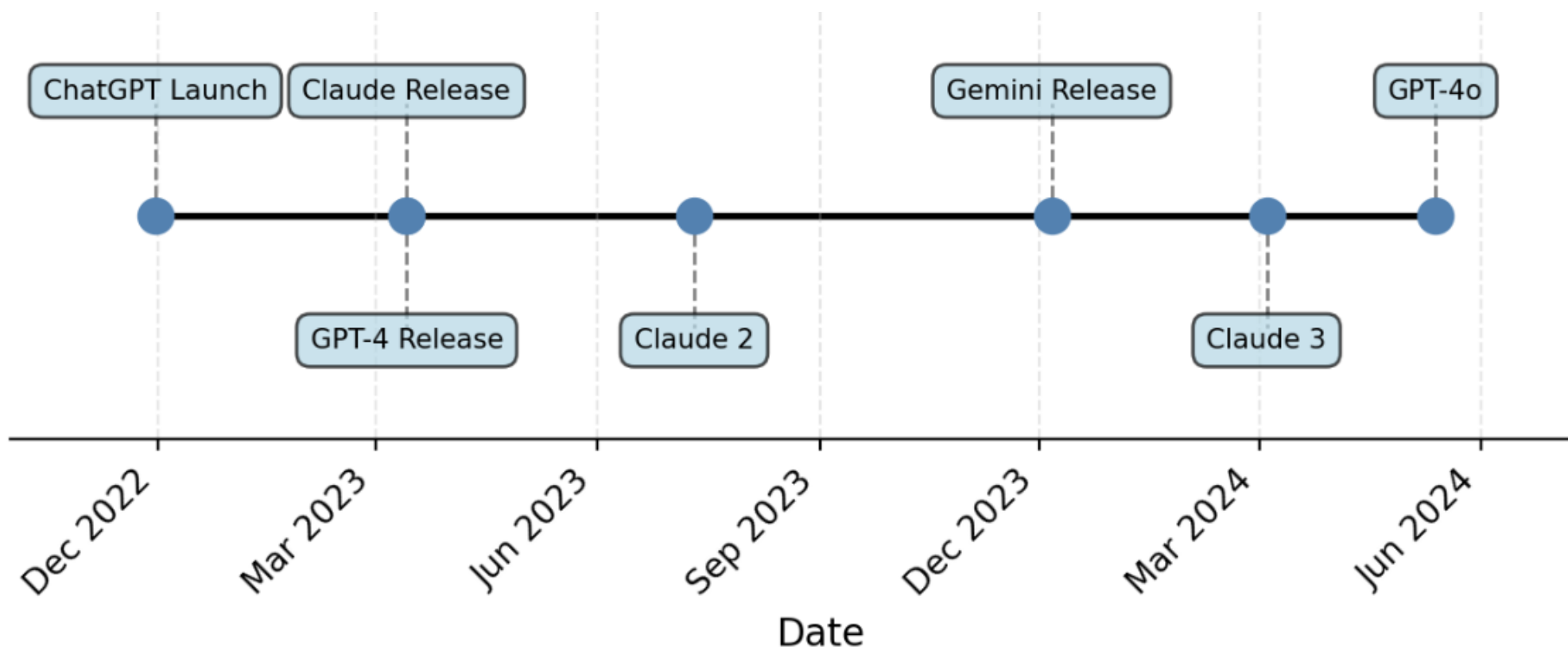
AI Approaches



Early work on AI impact on Labor Market

- We are going to explore very early work on how AI impacts the labor market
- Brynjolfsson, Chandar, and Chen ([2025](#)) use data from ADP (a payroll company) to understand how AI has impacted labor market
 - Most administrative datasets not available for such recent periods
 - Survey-based datasets (e.g. CPS) don't have quite enough observations to study impacts of automation across occupations

Timeline of LLMs



Measuring Exposure to AI

- Past work measures exposure to technology generally by studying tasks that individuals in different occupations perform and thinking through whether computers can also perform these task
 - This is a viable strategy for AI as well
- One alternative interesting approach has been to take information directly from the AI queries

Anthropic's Economic Index

- Anthropic took millions of conversations from Claude and studied what people generally were querying
- They then mapped these queries to tasks in O*NET
- An occupation is measured as exposed to AI if a lot of their tasks are associated with Claude Queries

Anthropic's Economic Index

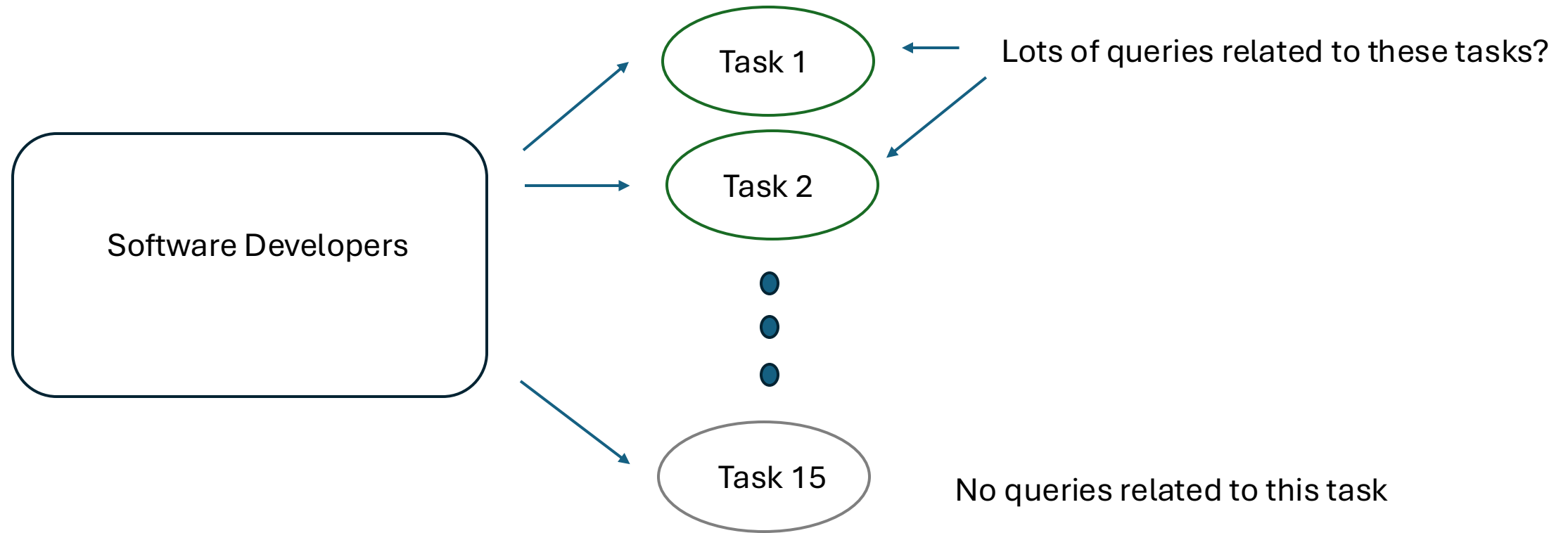
First step: Map every query to a task in O*NET

Query: Can you look through this code and see if there are any errors

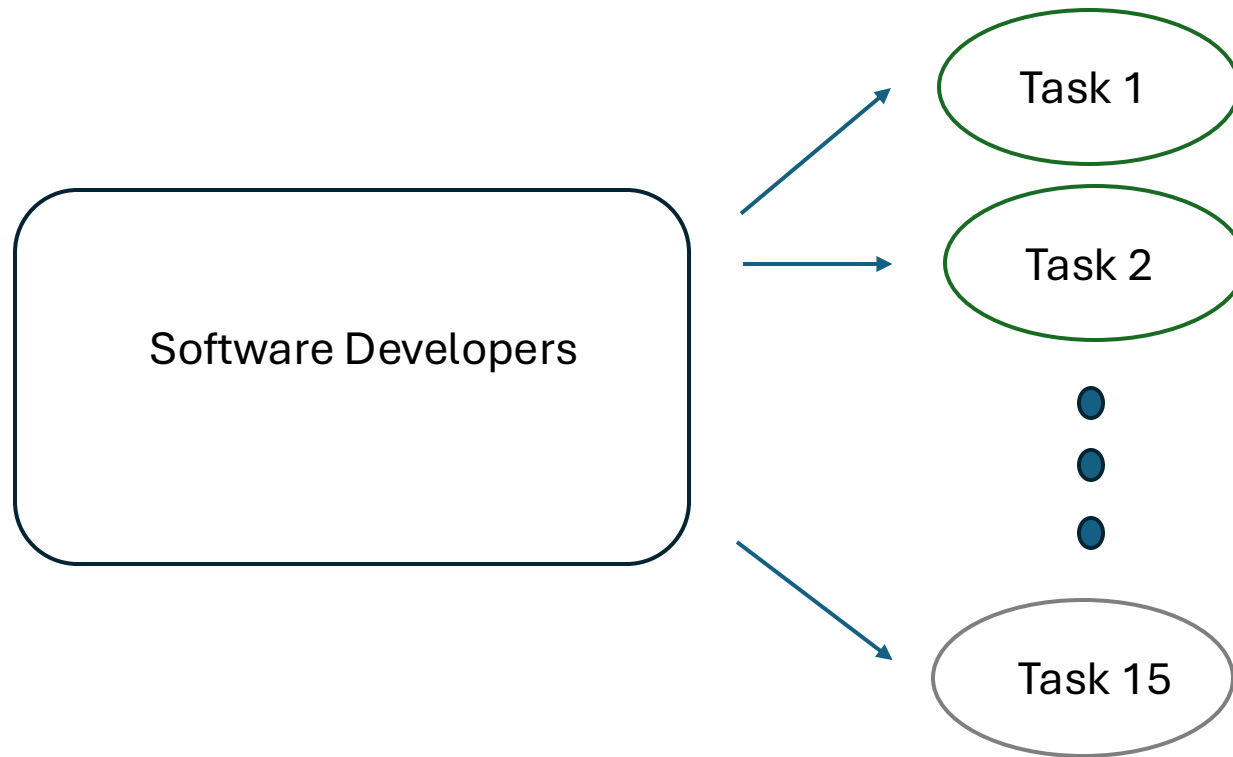


Task: modify existing software to correct errors, to adapt it to new hardware, or to upgrade interfaces and improve performance.

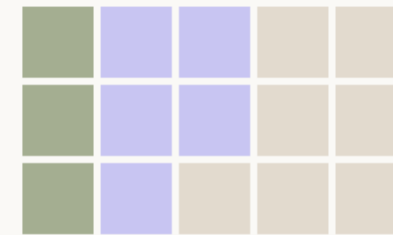
Anthropic's Economic Index



Anthropic's Economic Index



Software Developers, Applications
Computer and Mathematical



5.20% usage

Each box is a task: green is automated, blue is augmented and beige are no queries

Comparing Occupations

Software Developers, Applications

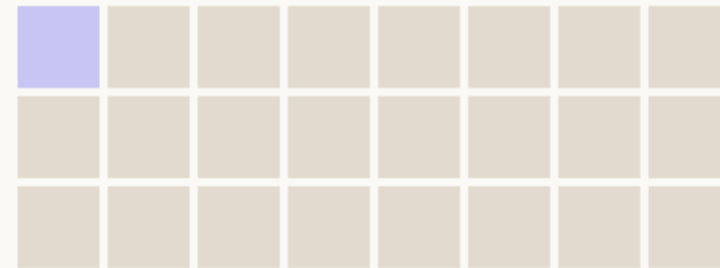
Computer and Mathematical



5.20% usage

Nursery Workers

Farming, Fishing, and Forestry

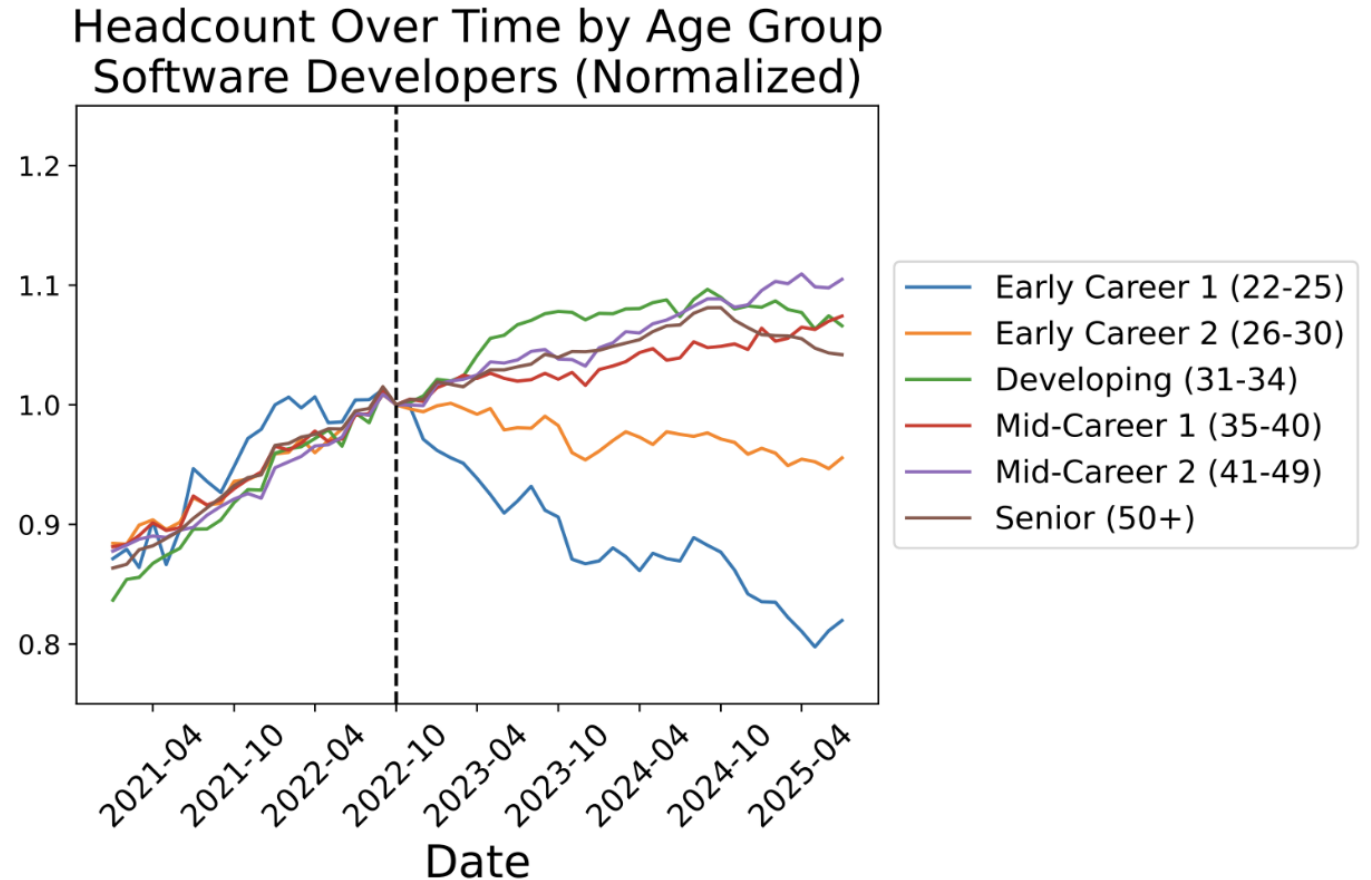


0.34% usage

Back to Brynjolfsson et al. (2025)

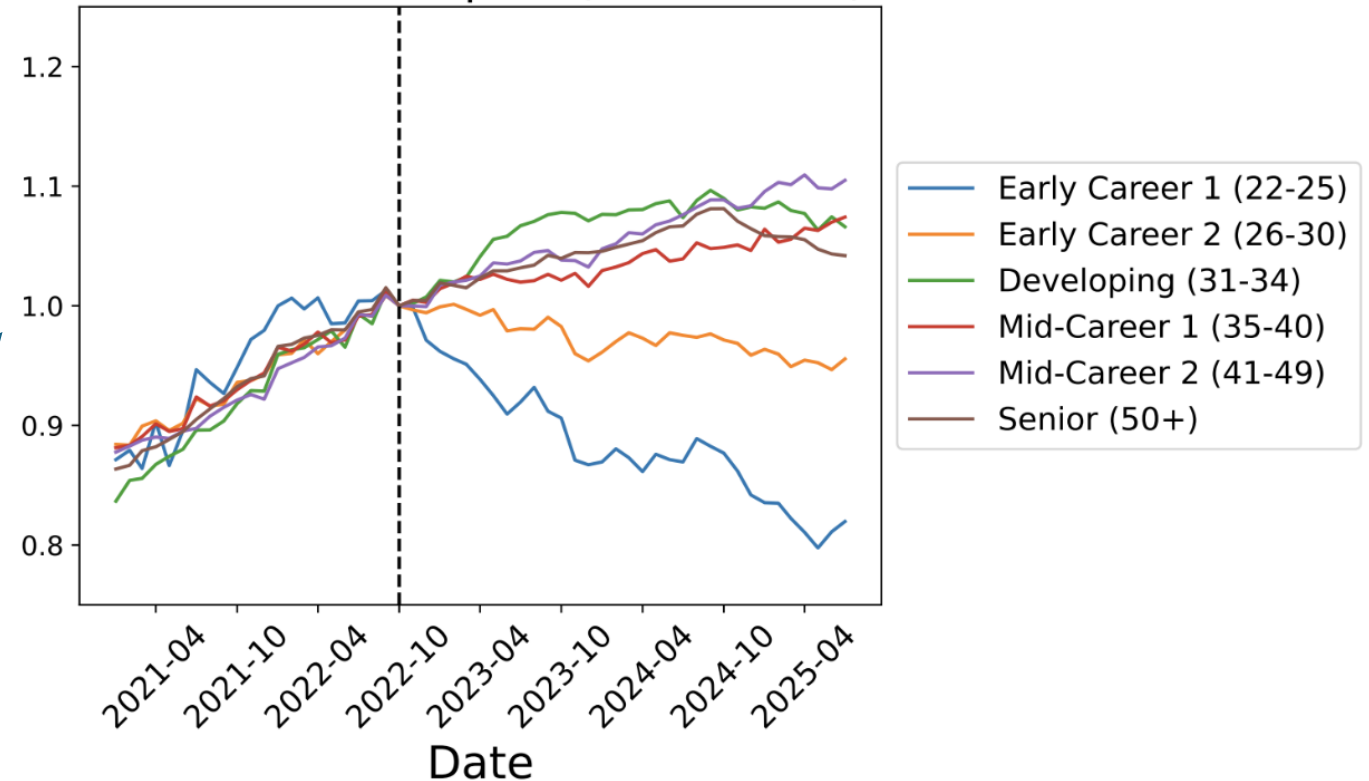
- First step: take some example cases, like software developers, that are intuitively highly exposed to AI
- Second step: see if patterns are consistent by degree of AI exposure
- **Key finding:** Age is important in understanding the impact of AI on the labor market (so far)

Employment of Software Developers over time



Employment of Software Developers over time

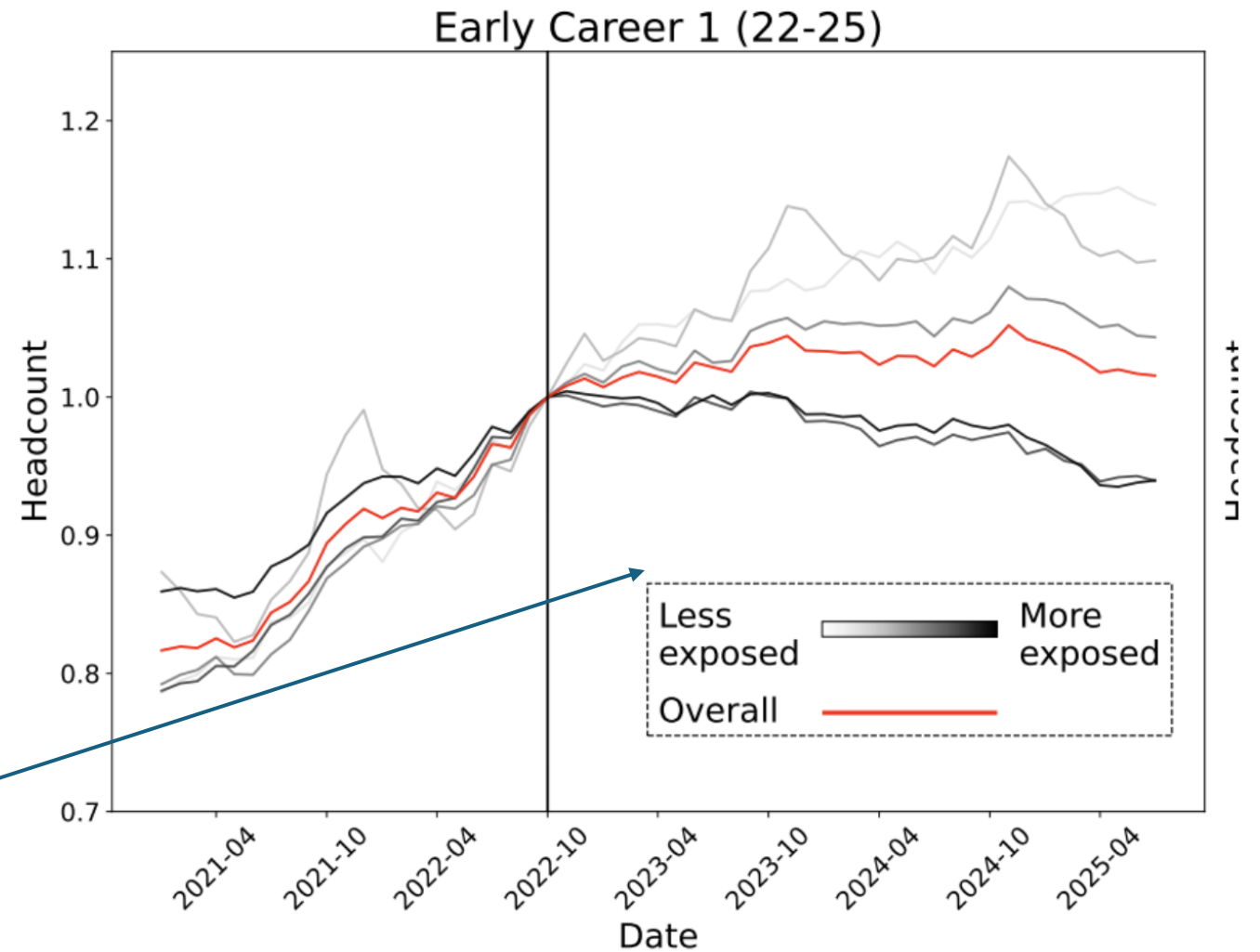
Headcount Over Time by Age Group
Software Developers (Normalized)



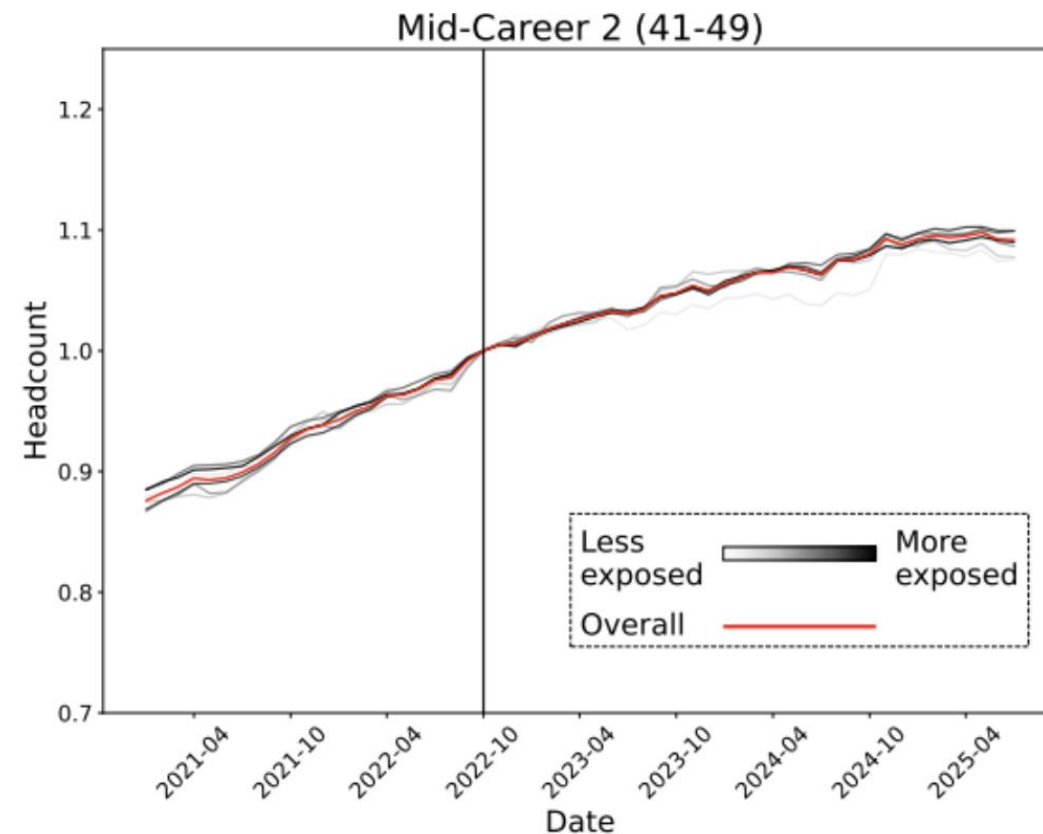
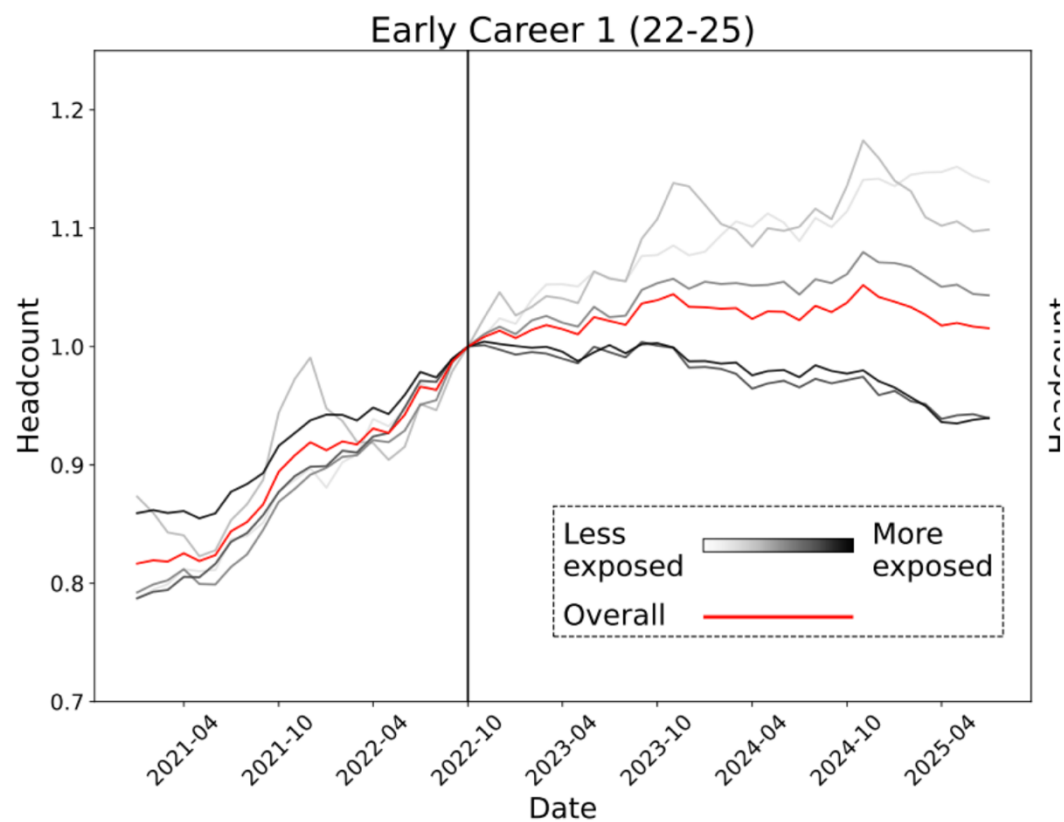
Employment is normalized to 1 at time October 2022. This means you can read y-axis as employment relative to October 22

Brynjolfsson et al. (2025)

Occupations broke into quintiles (5
groups) by level of AI exposure

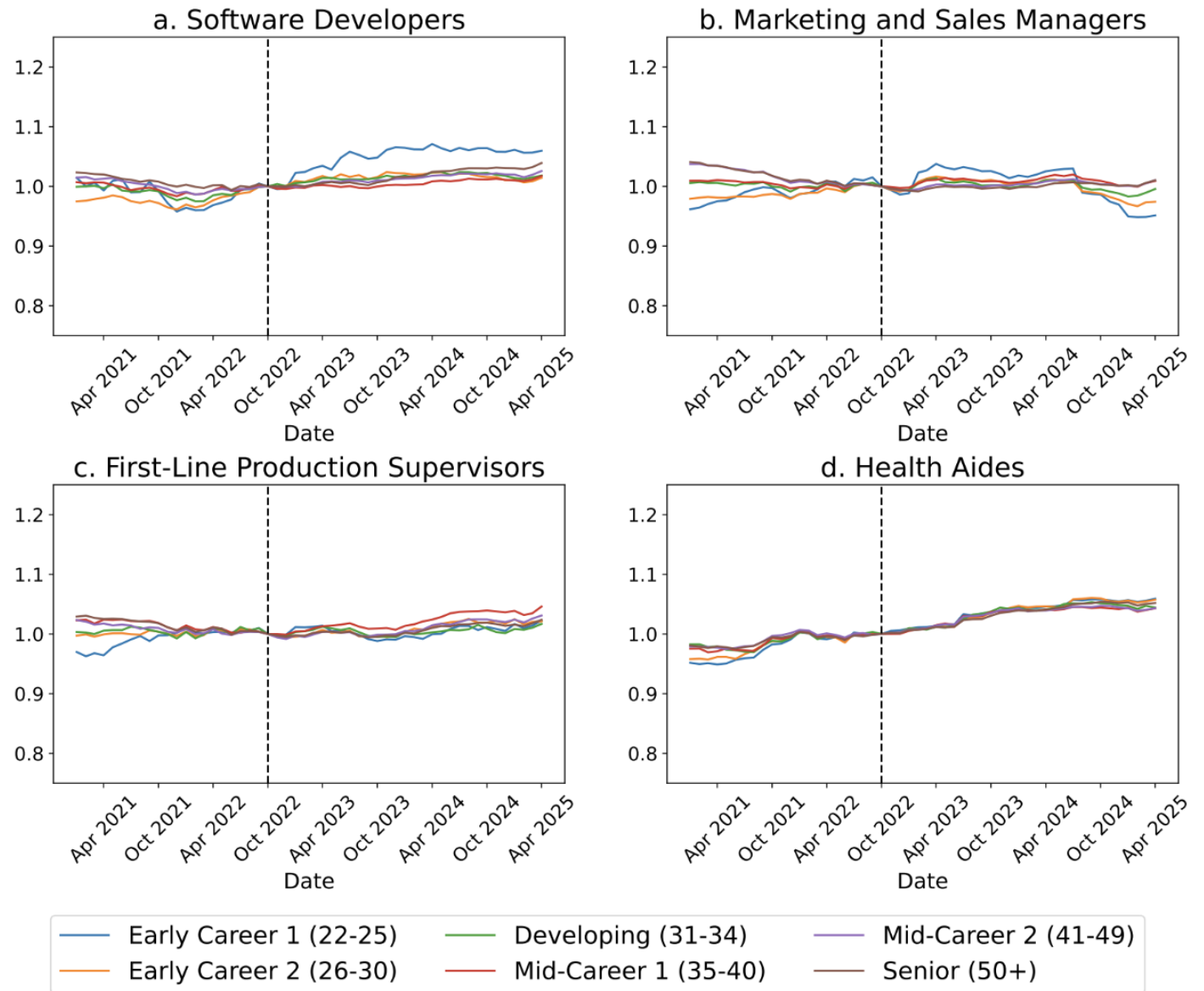


Brynjolfsson et al. (2025)



Brynjolfsson et al. (2025)

Outcome here is wages instead
of employment

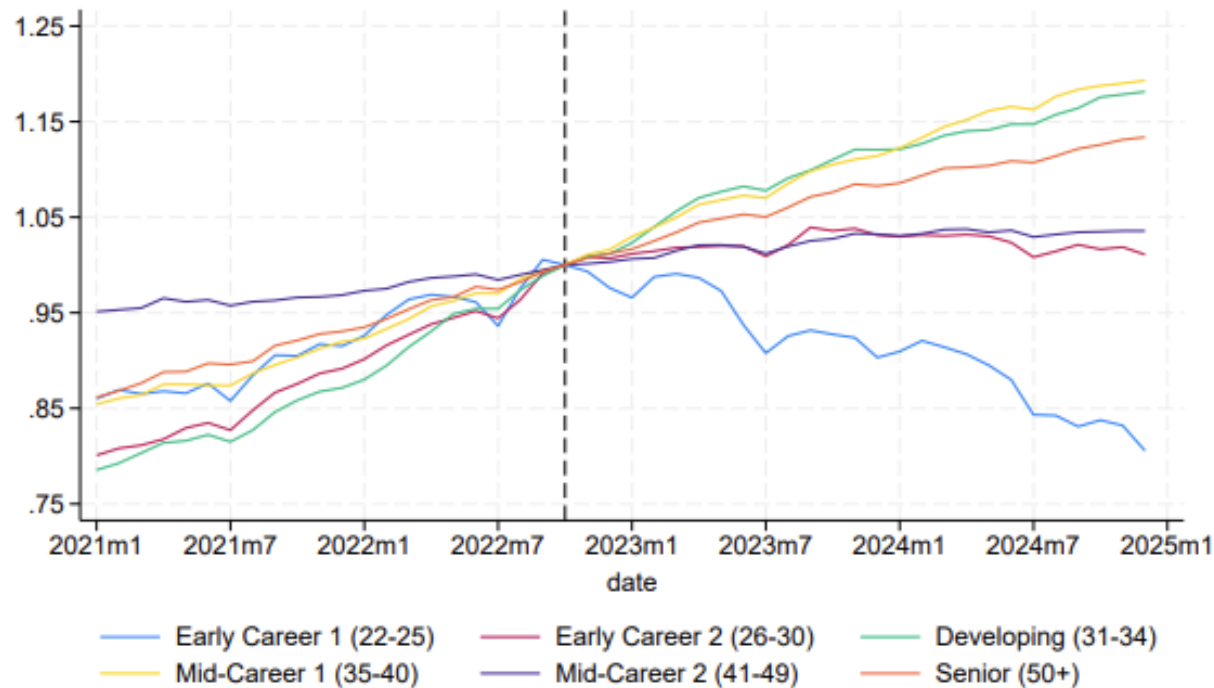


Humlum and Vestergaard (2025)

- Humlum and Vestergaard (2025) perform a similar study in Denmark
- **Key difference:** use worker survey to ask how their employers have responded to AI. They receive around 25,000 responses and can link this information to a number of firm-level outcomes
- This allows them to compare firms that adopt/encourage AI use vs. firms that don't adopt/encourage AI use

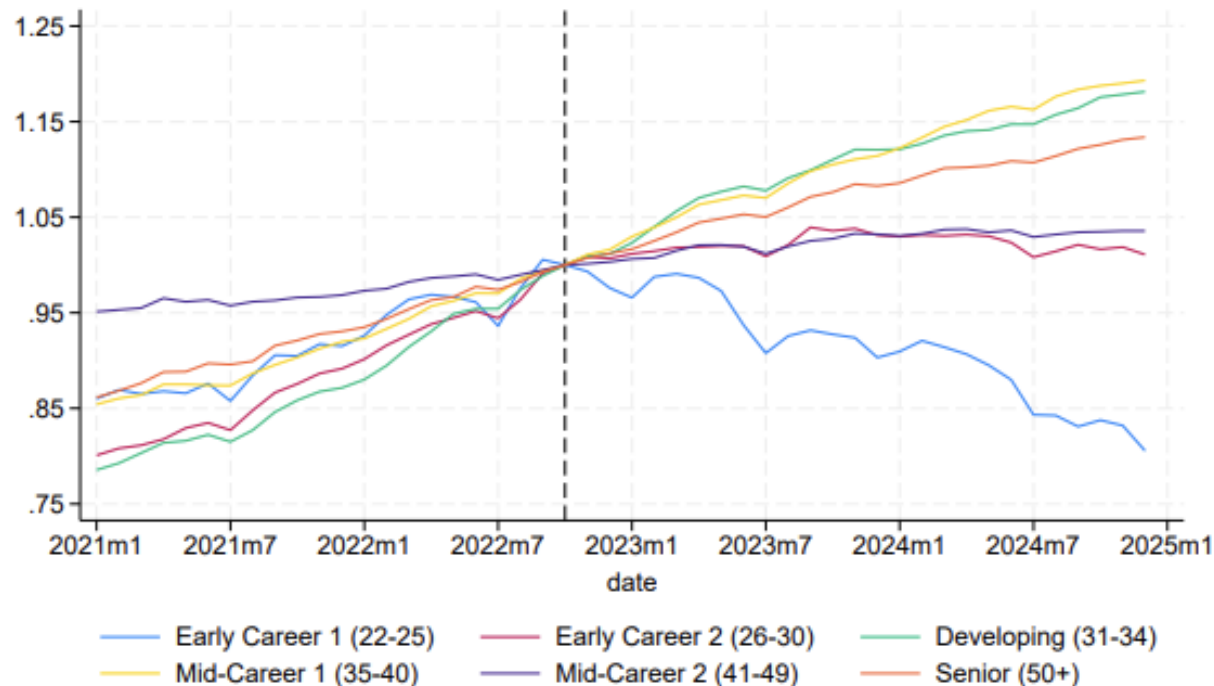
Humlum and Vestergaard (2025) find similar trends as Brynjolfsson et al. (2025)

(a) Software Developers



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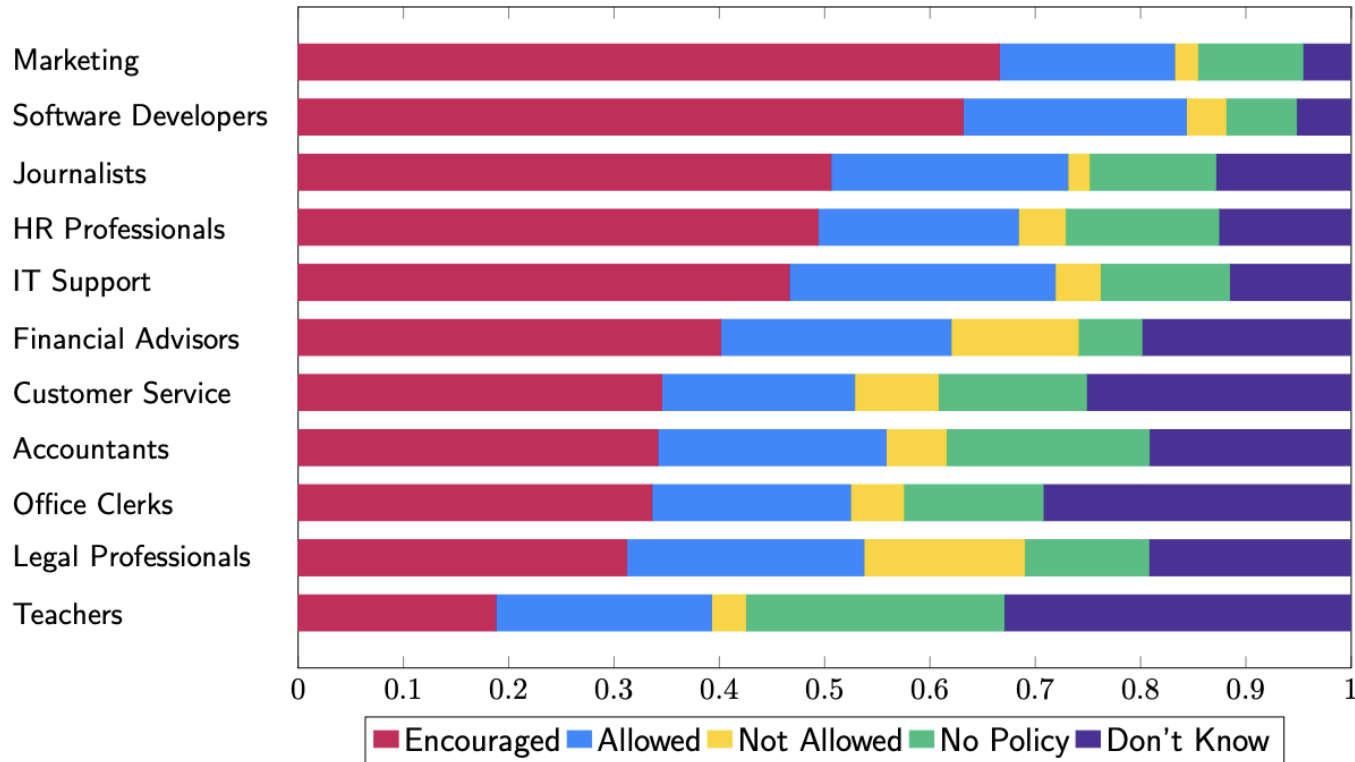
One concern with a plot like this is what if ChatGPT release happened to coincide with changes in labor market

It does appear like the impact is unexpectedly rapid

Humlum and Vestergaard study whether there are bigger employment declines at adopters vs. non-adopters

Firms vary in their response to AI

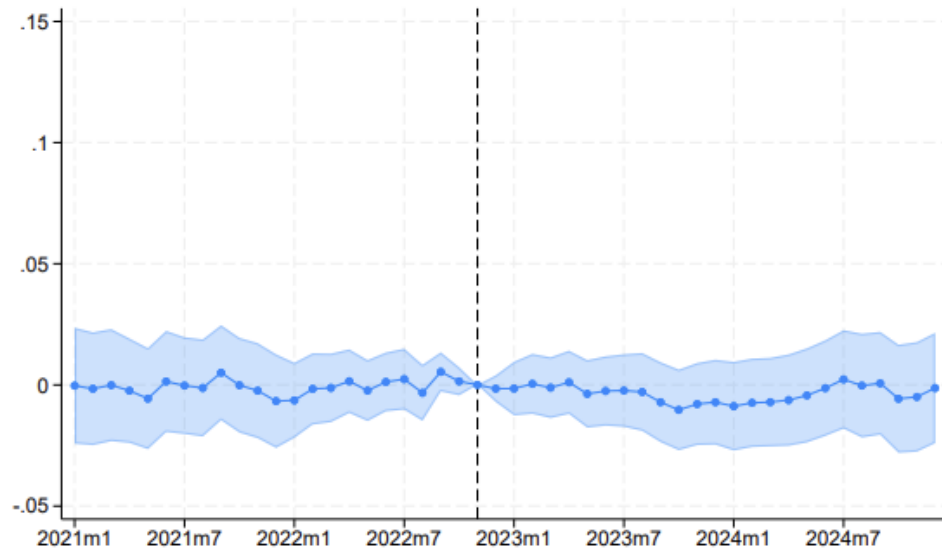
Figure 1: Employer Policies for AI Chatbot Use



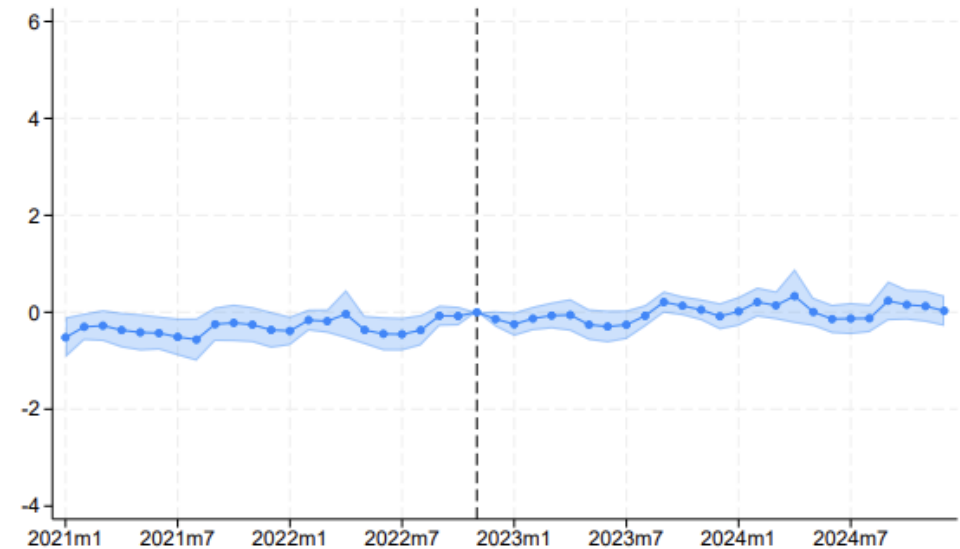
Humlum and Vestergaard (2025) find adopters and non-adopters have very similar employment levels

Figure 8: Labor Market Trends (Pre- and Post-Adoption)

(a) Log Total Employment



(b) Employment of Early-Career Workers



Suggestive evidence that maybe labor-market trends changed around the same time ChatGPT was adopted

Taking Stock

- Overall (across all ages) unemployment rates are still low



URATE:

**Entire
Population**

Taking Stock

- Initial evidence on AI does suggest younger workers are particularly impacted in certain occupation, but this is not yet obvious in aggregate statistics



URATE:

Ages 16-24

Looking Toward the Future

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- But this technology is very new and advancing quickly

Looking Toward the Future

- Initial evidence on AI does suggest younger workers are particularly impacted in certain occupations
- But this technology is very new and advancing quickly
- In the past, there is not a clear example of technology generating large and persistent increases in unemployment
 - This is despite it consistently being predicted, including by famous economists (e.g. Keynes, Leontief)
 - However, there isn't a clear "law" to this. It's possible future is different